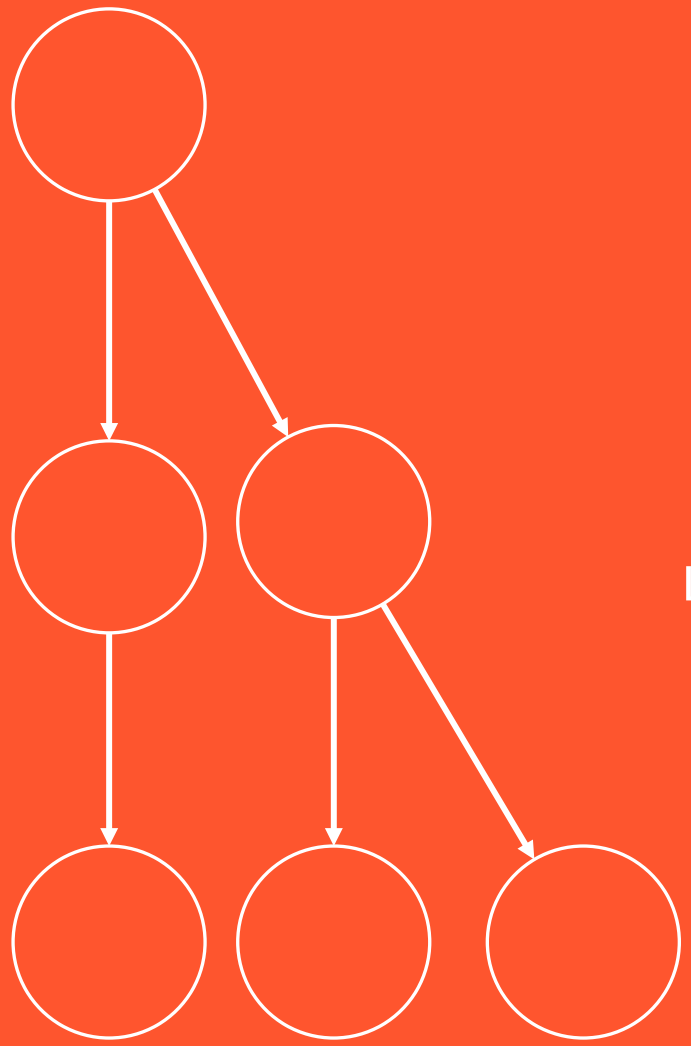


BCOG 100

Spatial Cognition

Lecture 9.1 — Finding the Way, and Deciding Where to Look

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Getting Back

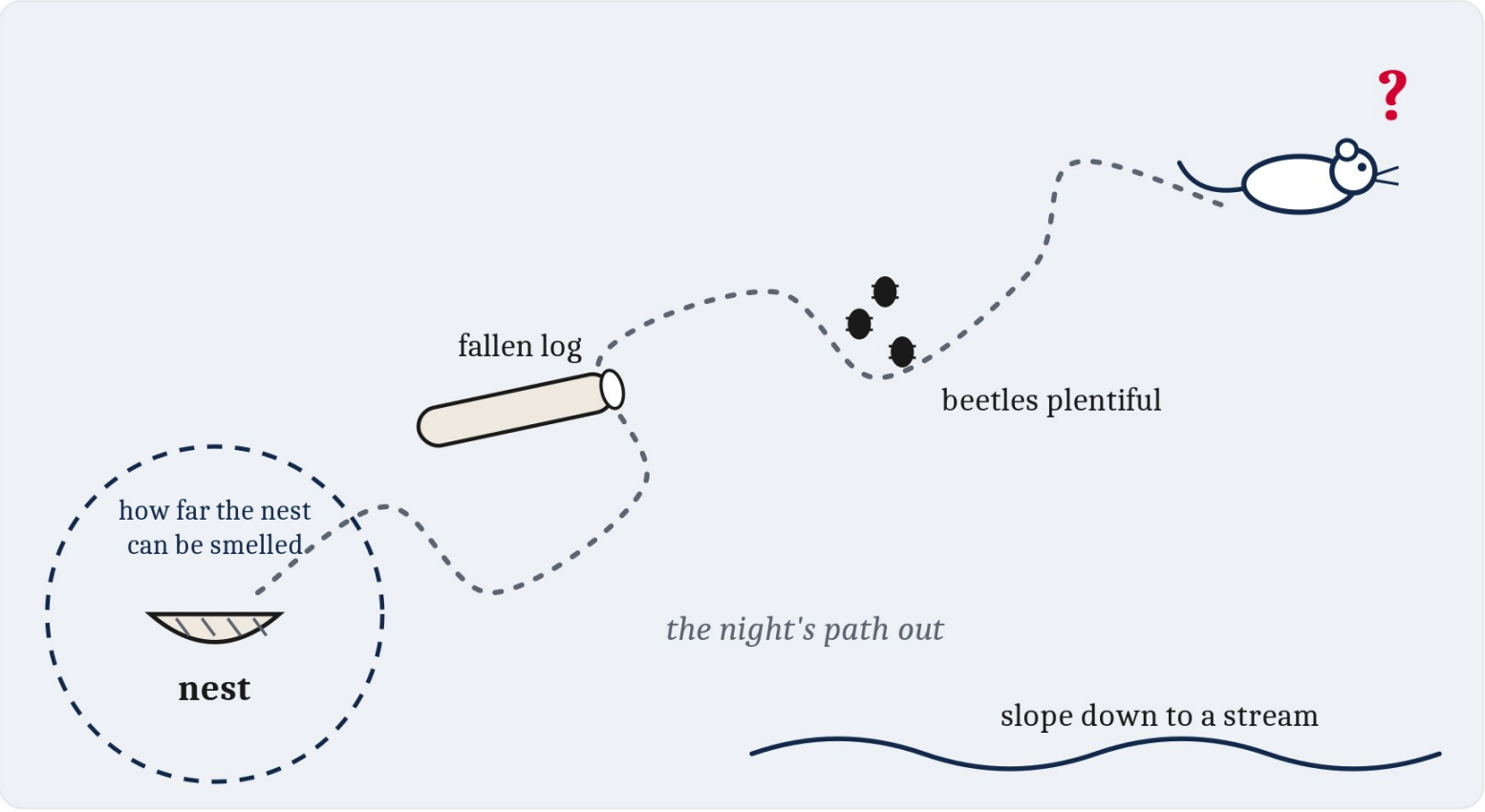
In your group, three minutes.

Your phone has died in a part of town you have never been to, and you need to get back to where you parked. Nothing at the car can be seen or heard from where you are.

What would you need to know to get back? What could you use?

Getting Home with Nothing to Steer By

Steering works only while the goal can be sensed. Far from its nest, an animal has two things left: signals of its own movement, and what it learned on earlier nights.



What it still has

Its own movement
how far, and in which direction,
it has gone
(vestibular system, proprioception)

What it learned on earlier nights
a fallen log, a slope down
to a stream, a spot where
beetles were plentiful

Nothing at the nest reaches the animal, so there is nothing to steer by.

A Small Warm Animal in the Dark

Early mammals were small, warm, and probably active at night or in dim light. A warm body needs a lot of food, so a lot of ground, and probably a way back home.



alongside the dinosaurs for well over 100 million years:
nearly all small insect-eaters, from shrew to badger size

A warm body needs fuel



Energy used over a whole day:
roughly ten times (bars drawn
to the reading's ratio)

Probably active at night, or in dim light

Converging evidence

Most living mammal lineages are active at night

Eyes suited to dim light, after two of four cone types were lost

Keener smell, high-pitched hearing, whiskers

Weak points

Some dinosaurs probably partly warm-blooded

Later predatory dinosaurs had eyes suited to the dark

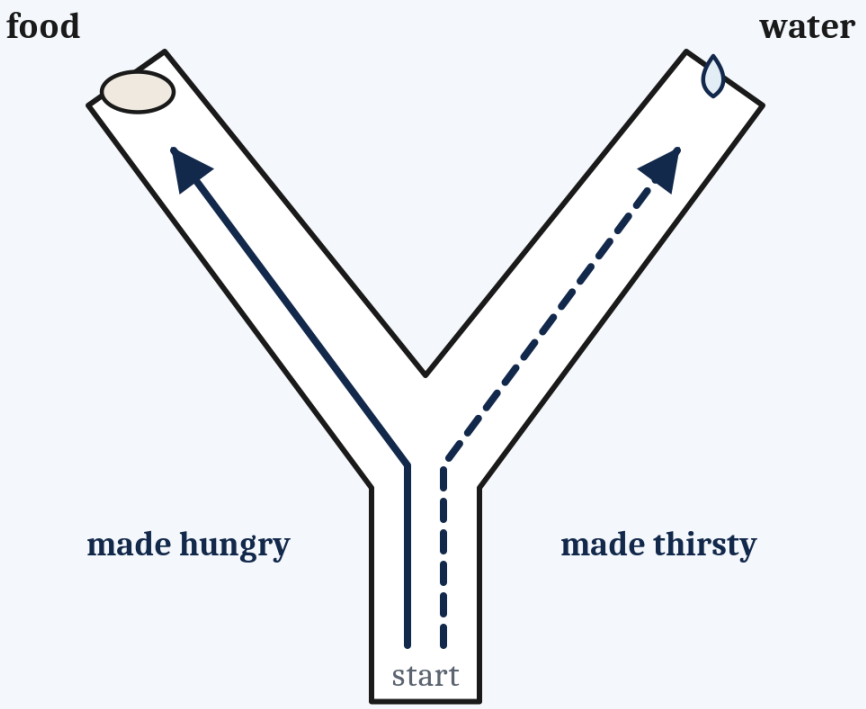
Lost cones fit dim light as well as full darkness

A reconstruction, from bones, genes, and living relatives.

What Has to Be Known

To reach a place it cannot sense, an animal needs to know four things. Which place is worth going to depends on the animal's state.

Trained neither hungry nor thirsty, then tested



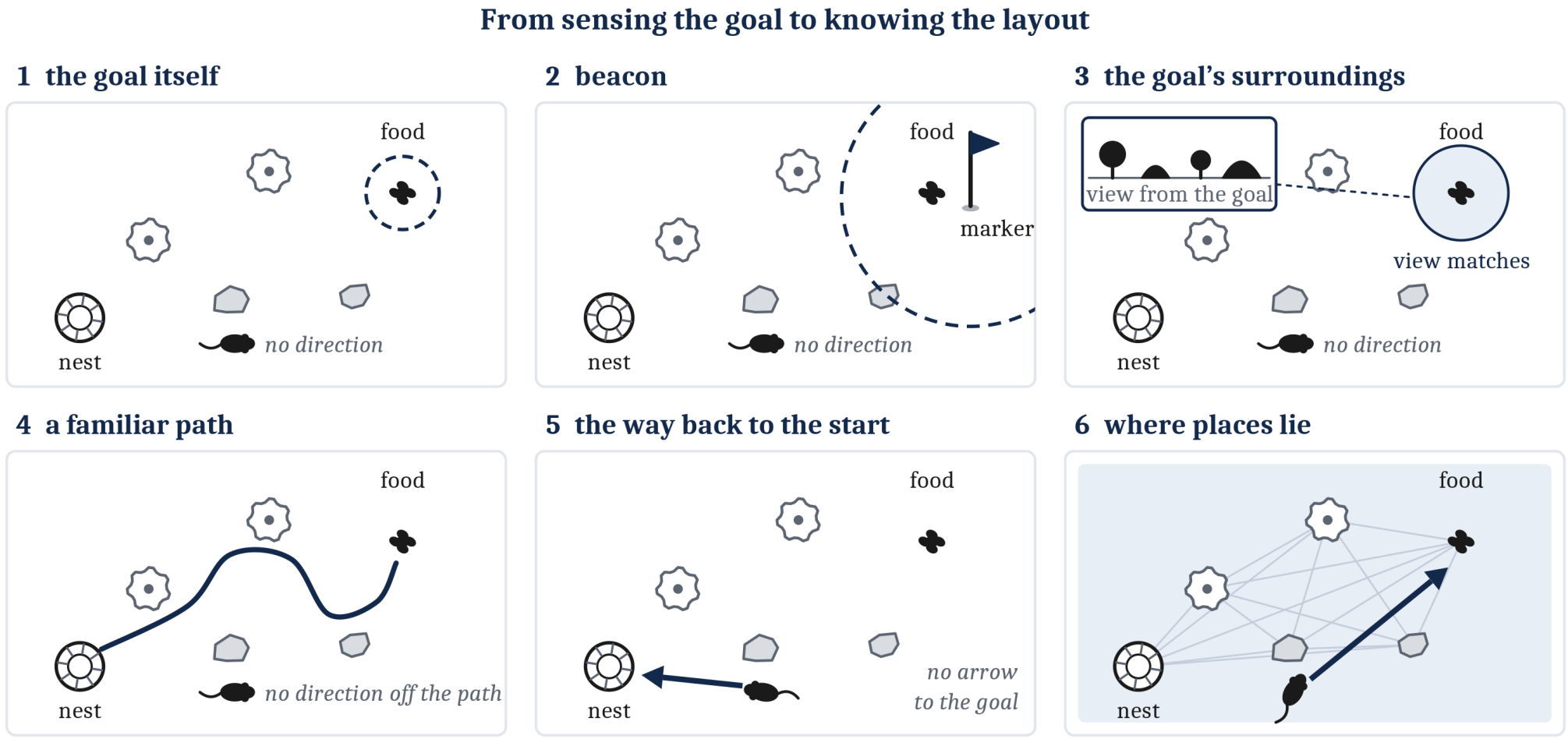
Four things to know

1. Which place is worth going to now — *set by the animal's state*
2. Where it is now
3. Where that place is
4. Which way to go

The rats had learned where things were; what they needed now picked where they went.

Six Kinds of Knowledge About a Goal

A **beacon** is something at or beside a goal that can be sensed from farther away. Only knowing where places lie relative to one another leads from a new start to any known place.



schematic; the same landscape in every panel, with the animal at a starting point it has never used

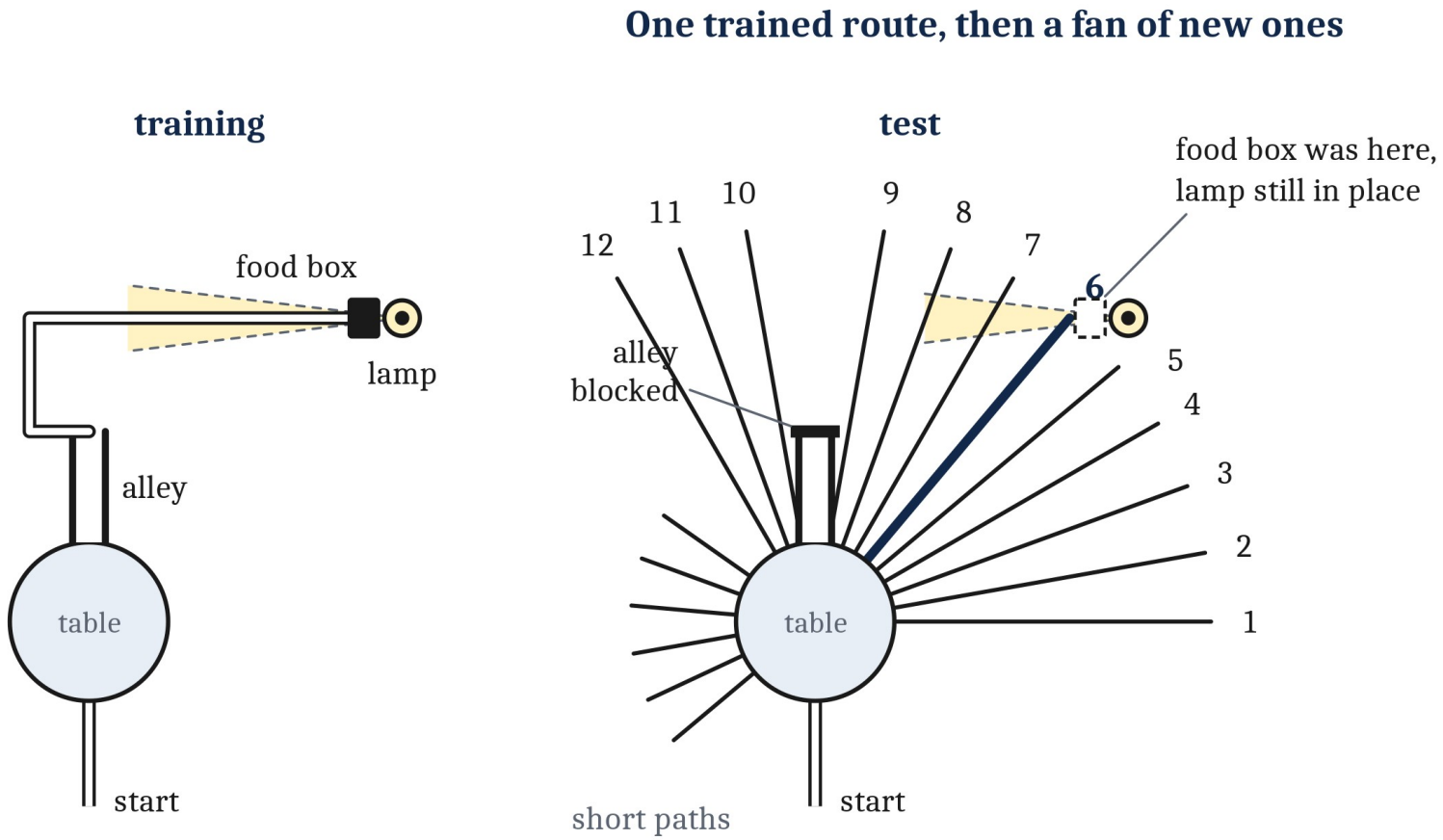
Cues, and How Each Fails Alone

A *cue* is something an animal's senses can detect as it moves; knowledge is what it has learned about how a cue relates to its goal. A **landmark** is a recognizable feature of the surroundings.

Cue	What it gives	How it fails alone
The animal's own movement	How far, and which way, it has gone	Small errors add up unseen; being carried registers as no movement; it points back only to the start
The sun	A direction	It moves through the day and the season; none at night
A nearby landmark	Position, closely, when near it	Changes with every step; may move, or look like another
A distant landmark	A direction	Indicates little about position
The shape of the surroundings	Which way the animal faces	A symmetric space gives two answers; open ground gives none
A gradient, such as a smell	The way toward its source	Runs out beyond its reach

What Do Animals Know? Tolman's Test

Latent learning is learning that does not show in behavior when it happens, and appears later, once there is a reason to use it. The test of knowing where places lie is a route never taken, but a new route alone is not proof.

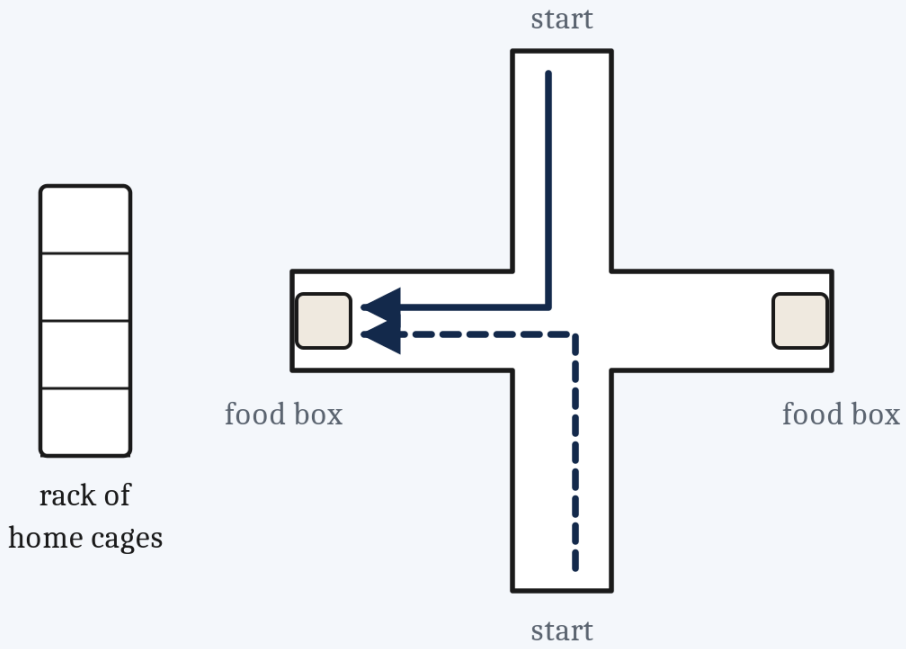


from above, simplified; paths 1 to 12 lie 10 degrees apart; redrawn after Tolman, Ritchie, and Kalish (1946)

Place Learning and Response Learning

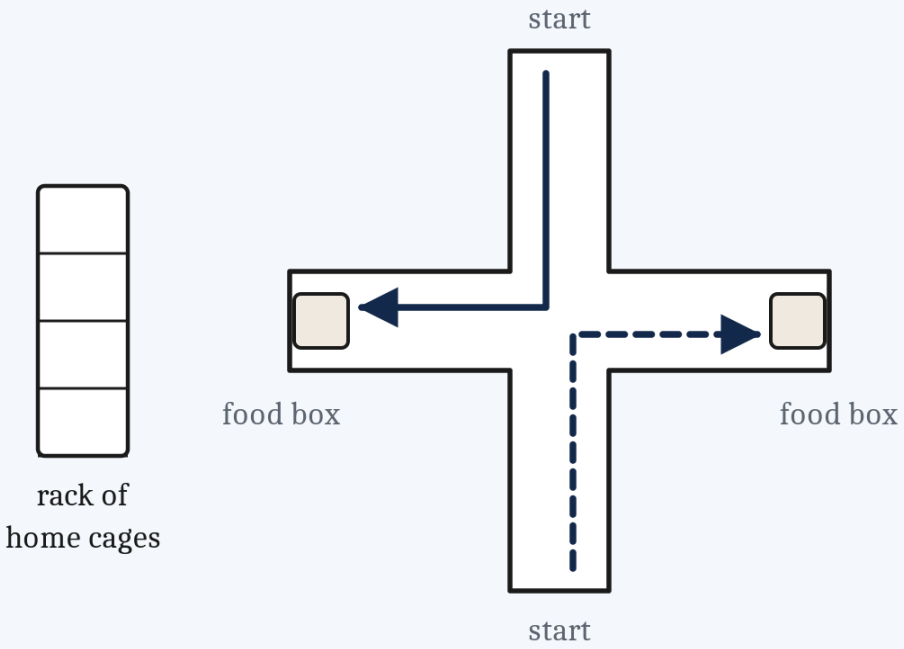
Place learning: going to a location, whatever movements it takes. **Response learning**: repeating a movement, whatever location it leads to. Which one rats show depends on the room.

Place learners



Always the same food box.
All 8 learned within 8 trials.

Response learners



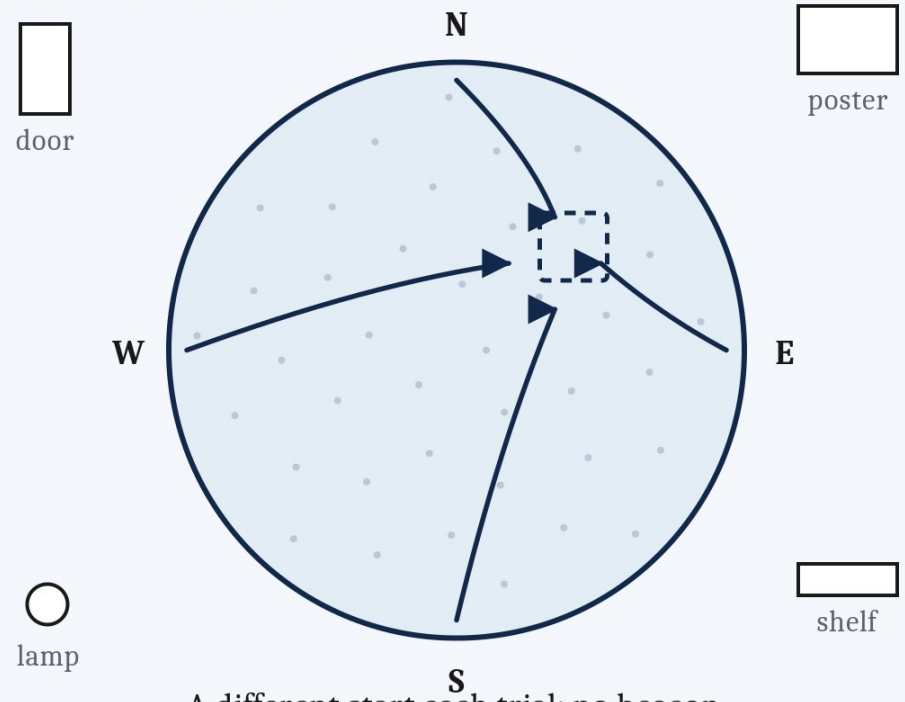
Always turn right.
Only 3 of 8 learned within 72 trials;
the rest kept going to one end of the room.

Distinct surroundings beyond the arms: rats went to the place. A plain enclosure: they repeated the turn.

Routes Never Taken

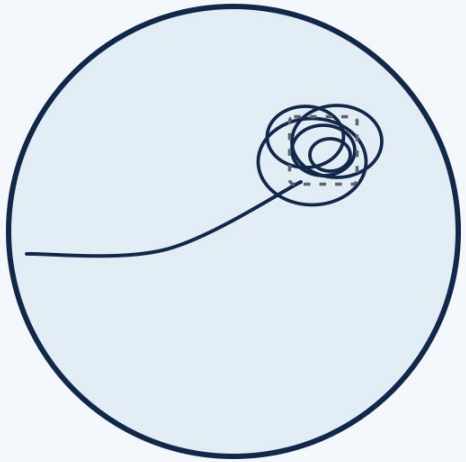
With better controls, animals can find a goal that nothing at the goal marks, and some animals take routes they have truly never taken.

The water maze



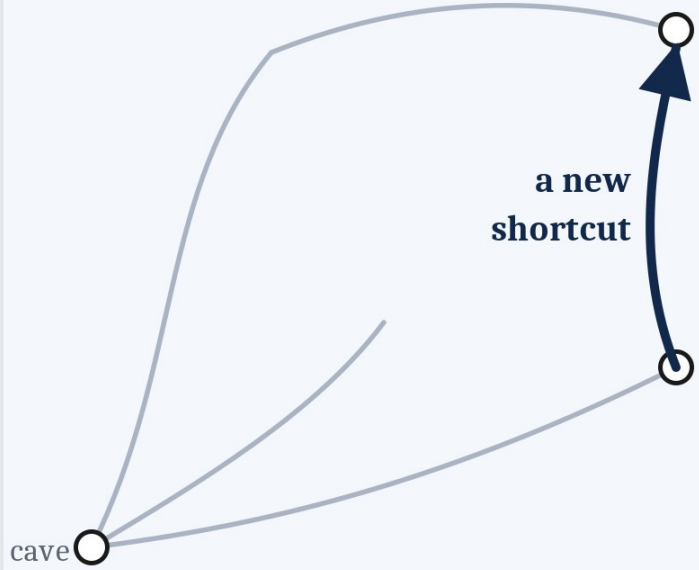
A different start each trial: no beacon, no fixed sequence of turns

Platform removed



Rats searched where the platform had been

Fruit bats tracked by GPS



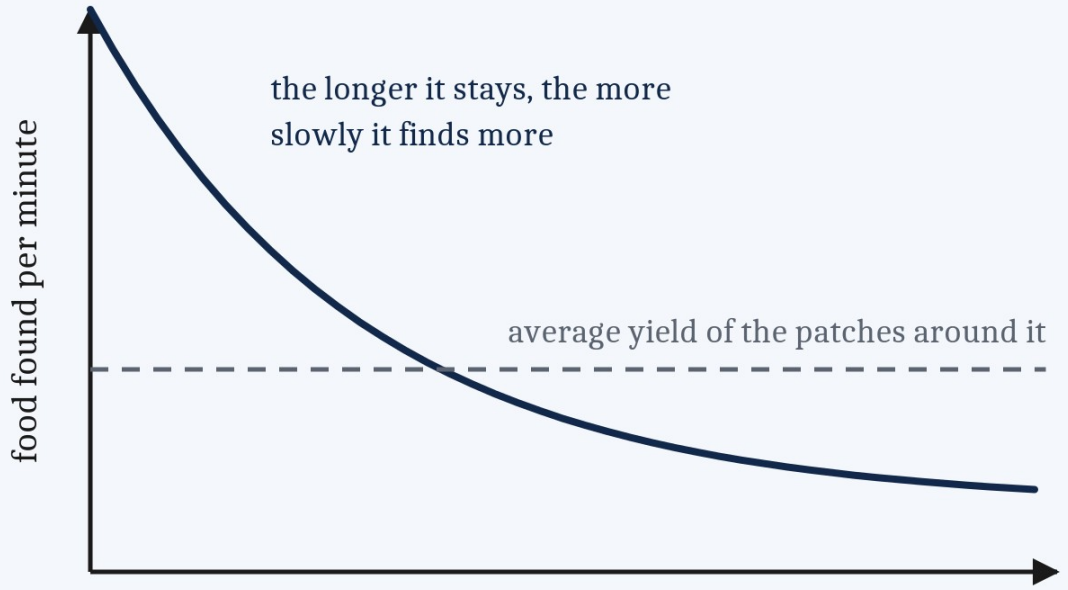
Gray: every route flown since the first flight. At least half of the shortcut lay more than 100 m from anywhere flown.

Animals can find a goal nothing at the goal marks, and some take routes truly never taken.

Whether to Keep Exploring

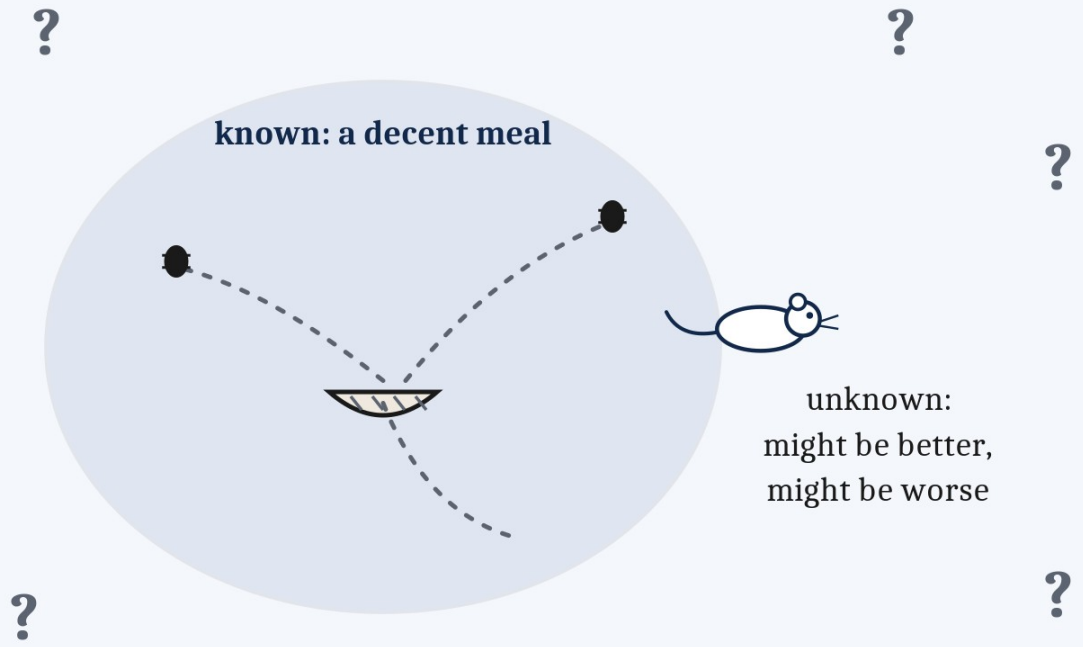
Exploration versus exploitation is the trade-off between taking an option already known to be decent, and sampling one that might be better and might be worse. Exploring is worth doing only because of the future.

Leave a patch that is running down?



Moving to another patch takes time.

Go somewhere never visited?



Choosing without knowing what is there.

Exploring is worth doing only because of the future.

Wanting to Know

Animals also seek information that no reward depends on. Not knowing comes in two kinds: a known question, and an open-ended one.

Information no reward depends on

Rats With nothing to do, spend more time in the unfamiliar parts of a maze

Monkeys Give up about a quarter of a reward's value to learn early whether it is coming

Pigeons

averages 2 pellets, with a signal of what is coming	averages 3 pellets, no signal

pigeons choose this one

Why accept less for information that cannot be used?

Two kinds of not knowing

A known question



Where is my car?
I will know it
when I see it.

An open-ended one



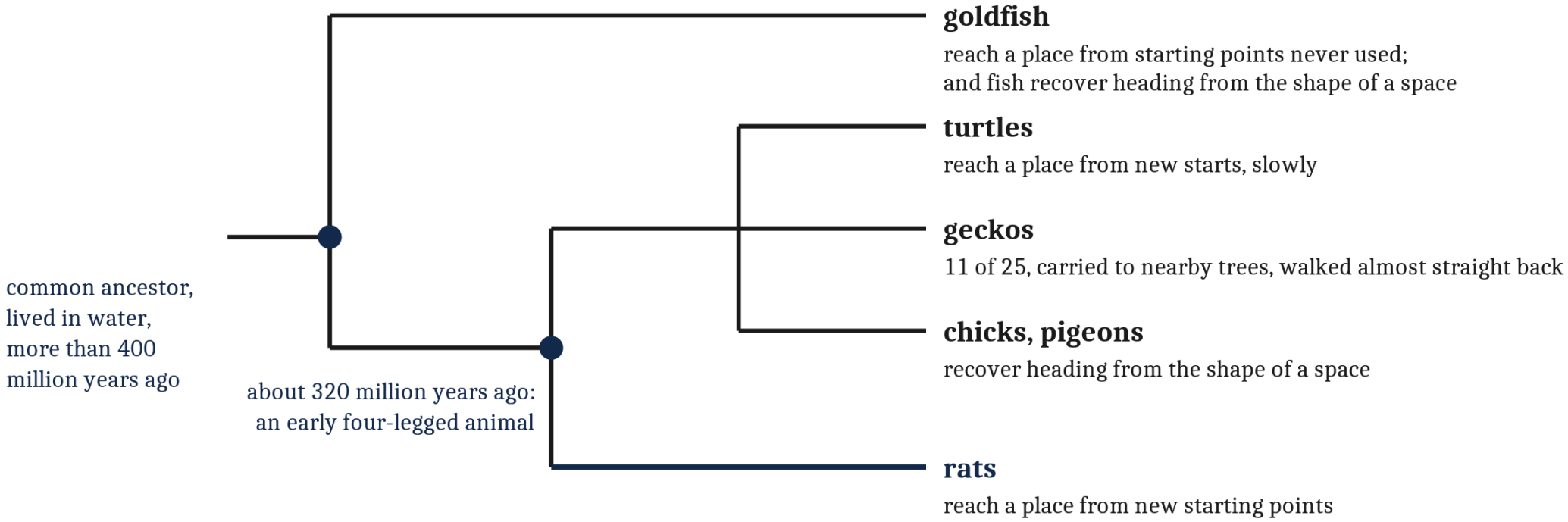
Which streets in a
new city will matter?
Not yet known what
is worth asking.

Monkeys and pigeons faced the first kind; an animal past the edge of known ground, the second.

What the Ancestors Already Had

Homology is a resemblance between two species that both inherited from a common ancestor. Reaching a place from a new start is very probably more than 400 million years old.

What living members of each group do



Insects keep track of the way back to the start and follow familiar views and paths, but with brain parts that are not counterparts of the vertebrates': likely convergent.

Which Kind of Knowledge?

In your group, four minutes. For each case, which kinds of knowledge could explain it? What test would tell them apart?

1. A dog lost three miles from home turns up at its own door.
2. A commuter who drives the same way every day gets lost when a road is closed.
3. A person who always navigates by phone cannot point toward home.

Image Sources (1 of 2)

Slide	Image	File	License	Source
3	Far from home, with nothing to steer by	fig-no-signal.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
4	The early mammal, as we reconstruct it	fig-early-mammal.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
5	Four things to know, and the state that picks the goal	fig-four-things.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
6	Six kinds of knowledge about a goal	fig-what-is-carried.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading · own work
8	The radiating-paths test	fig-radiating-paths.png	CC BY 4.0	Jon Willits — drawn for the Module 9 reading, redrawn from Tolman, Ritchie, and Kalish (1946) · own work
9	The plus-shaped maze	fig-plus-maze.png	CC BY 4.0	Jon Willits — drawn for this lecture, after Tolman, Ritchie, and Kalish (1946) · own work
10	Closing the gaps	fig-water-maze.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
11	The exploration problem, in two forms	fig-two-forms.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

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12	Information for its own sake, and two kinds of not knowing	fig-wanting-to-know.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	Spatial abilities shared across vertebrates	fig-ancestors.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work